9.7 Equal Sums of Powers

WORKSHEET

The question

STEP I 1992, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If a polynomial $\mathrm{p}(x) = \sum c_k x^k$ has $\mathrm{p}(1) = 0$, what does that say about the coefficients? What about $\mathrm{p}'(1) = 0$?

STAGE 1 A triple root

Ask yourself Factorise $1 - x^2$ and $1 - x^4$. How many factors of $1 - x$ appear?

STAGE 2 Derivatives at 1

Ask yourself If $(1-x)^3$ divides p_0, what power of $(1-x)$ divides p_0'? And then $\mathrm{p}_1 = x\mathrm{p}_0'$ and p_1'?

STAGE 3 Reading off the coefficients

Ask yourself What are $\mathrm{p}_0'(1)$ and $\mathrm{p}_1'(1)$ in terms of the coefficients c_k?

STAGE 4 One more factor

Pause and try To make the cubes balance too, we need a fourth factor of $(1-x)$. Which polynomial should replace p_0?

Try this next

1. **Shifting.** Try STEP II 2025, Question 8 (in OCR's download of the 2025 papers, www.ocr.org.uk/Images/750486-2025-past-papers.zip). It shows how to turn a split that balances sums and squares into a larger one that balances cubes as well, by shifting one list and combining.

2. **Squares of squares.** (Also from STEP II, 2025.) The sets $A = \{1,3,4,5,9,11\}$ and $B = \{2,6,7,8,10\}$ have equal sums and equal sums of squares. Split $\{n^2, (n+1)^2, \ldots, (n+11)^2\}$ into two sets with equal sums, for every positive integer n. (*Hint:* add 0 to the second set.)

3. **Translation invariance.** Show that if two sets of numbers have equal sums of rth powers for $r = 0, 1, \ldots, k$, then so do the sets obtained by adding the same number c to every element. Why does this mean that the question's sets $\{1, \ldots, 15\}$ could

equally be replaced by any 16 consecutive integers?

4 Fair sweets. Sweets have values $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{8}$. Compare the totals A gets with the orders $ABABABAB$ and $ABBABAAB$.

Further reading

- **Jean-Paul Allouche & Jeffrey Shallit, "The ubiquitous Prouhet–Thue–Morse sequence"**, in *Sequences and Their Applications* (Springer, 1999). A survey of the sequence's many appearances; free online.
- **Matt Parker, *Things to Make and Do in the Fourth Dimension*** (Penguin). Includes equal power sums and other number curiosities.
- **Joshua Cooper & Aaron Dutle, "Greedy Galois games"**, *American Mathematical Monthly* 120 (2013). Fair turn-taking and Thue–Morse.

The full guided solution, and what the question leads to at university, is in Section 9.7:
https://beyondstep.keithmcnulty.org/ch09/equal-power-sums.html